

12.2 Centripetal Acceleration

Question Paper

Course	CIE A Level Physics
Section	12. Motion in a Circle
Topic	12.2 Centripetal Acceleration
Difficulty	Easy

Time allowed: 40

Score: /33

Percentage: /100

Question 1a

A centripetal force causes a centripetal acceleration.

State **two** properties of the centripetal force that is required for this.

[2 marks]

Question 1b

State the equation for centripetal acceleration.

[1]

State the definition and unit of each variable in part (i)

[3]

[4 marks]

Question 1c

Calculate the angular speed of a satellite in a geostationary orbit.

[3 marks]

Question 1d

The satellite has an altitude of 36 000 km.

Calculate the centripetal acceleration of the satellite.

Radius of Earth = 6400 km

[3 marks]

Question 2a

State what is meant by *centripetal force*.

[2 marks]

Question 2b

State the centripetal force in each of the following situations

(i)

Venus orbiting the Sun

[1]

(ii)

a car driving around a roundabout

[1]

(iii)

rotating a ball tied to the end of a piece of string.

[1]

[3 marks]

Question 2c

The Moon orbits the Earth in a circular path.

Explain, with reference to Fig 1.1, why the path of the moon is circular.

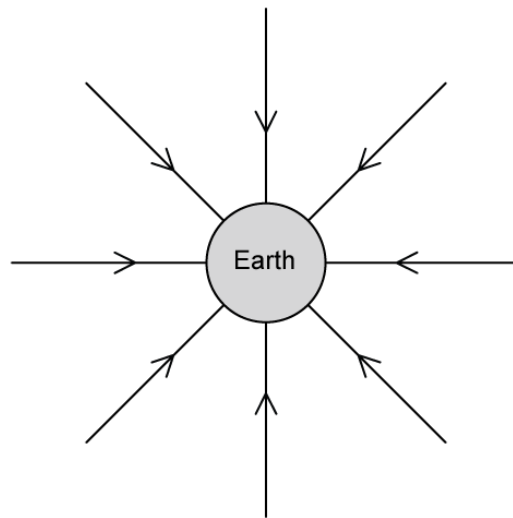


Fig 1.1

[2 marks]

Question 2d

The centripetal force of the Moon is $2.21 \times 10^{20} \text{ N}$.

Calculate the distance of the Moon from the surface of the Earth.

Mass of the Moon = $7.35 \times 10^{22} \text{ kg}$

Radius of Earth = $6.4 \times 10^6 \text{ m}$

Linear speed of the Moon = 1023 m s^{-1}

[4 marks]

Question 3a

Two of Jupiter's moons, Ganymede and Callisto orbit around Jupiter in a circular path, as shown in Fig 1.1.

Fig 1.1 shows the North pole of Jupiter, where both moons orbit counter-clockwise.

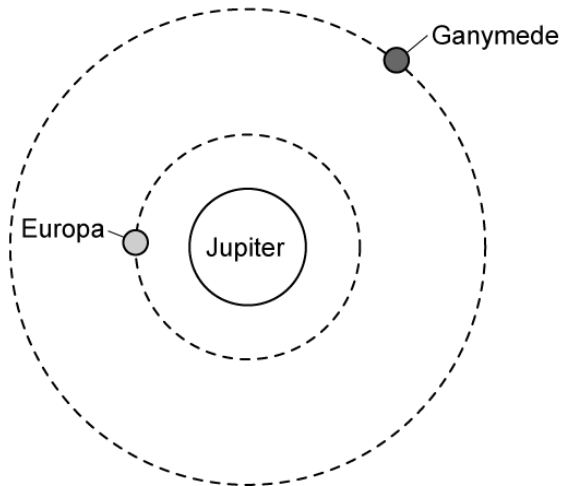


Fig 1.1

Identify, by drawing arrows, the following:

- (i)
the centripetal force on each moon. Label this F .

[2]

- (ii)
the linear velocity of each moon. Label this v .

[2]

[4 marks]

Question 3b

Both of Jupiter's moons travel at a speed of roughly $10\,000\text{ m s}^{-1}$.

State which moon will have a longer time period. Explain your answer.

[3 marks]

Question 3c

The centripetal acceleration of Ganymede is 0.11 m s^{-2} .

The mass of Ganymede is $1.48 \times 10^{23}\text{ kg}$.

Determine the centripetal force on Ganymede.

[2 marks]

Question 3d

Explain how the centripetal force of an object with the same mass and orbit as Ganymede would compare if it travelled with twice the angular speed.

[1 mark]

Model Answers: Easy

1a

(a) Two properties of the centripetal force that cause a centripetal acceleration are:

- (It must be) constant; [1 mark]
- It must be perpendicular to the direction of motion **OR** towards the centre of the circle; [1 mark]

[Total: 2 marks]

1b

(b)

(i) The equation for centripetal acceleration is:

- $a = r\omega^2$ **OR** $a = \frac{v^2}{r}$ [1 mark]

(ii) The definition and unit for each variable are:

- a is (centripetal) acceleration **AND** in m s^{-2} [1 mark]
- r is the radius of the (circular) orbit **AND** in m [1 mark]
- v is the linear speed **AND** in m s^{-1} [1 mark]

[Total: 4 marks]

1c

(c) The angular speed of an object in a geostationary orbit is:

Determine the time period of the satellite:

- A geostationary orbit has a time period 24 hours (1 day)

- 24 hours = $24 \times 60 \times 60 = 86\,400 \text{ s}$ [1 mark]

Determine the equation for the angular speed:

- $\omega = 2\pi f = \frac{2\pi}{T}$

Substitute in the values:

- $\omega = \frac{2\pi}{86\,400}$ [1 mark]
- $\omega = 7.272 \times 10^{-5} = 7.3 \times 10^{-5} \text{ rad s}^{-1}$ [1 mark]

[Total: 3 marks]

1d

(d) The centripetal acceleration of the satellite is:

List the known quantities:

- Angular speed of the satellite, $\omega = 7.272 \times 10^{-5} \text{ rad s}^{-1}$
- Radius of Earth, $R = 6400 \text{ km} = 6400 \times 10^3 \text{ m}$
- Altitude of the satellite, $h = 36\,000 \text{ km} = 36\,000 \times 10^3 \text{ m}$

Determine the radius of the satellite's circular orbit, r .

- $r = R + h = (6400 \times 10^3) + (36\,000 \times 10^3) = 42.4 \times 10^6 \text{ m}$ [1 mark]

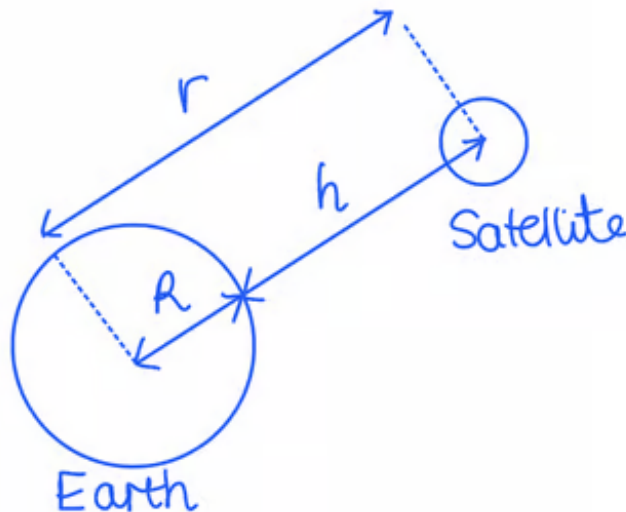
Substitute values into the centripetal acceleration equation, a :

- $a = r\omega^2 = (42.4 \times 10^6) \times (7.272 \times 10^{-5})^2$ [1 mark]
- $a = 0.2242 = 0.22 \text{ m s}^{-2}$ [1 mark]

[Total: 3 marks]

Always make sure you use the correct value for the radius of orbit, r . This is the distance from the **centre** of the two bodies.

Therefore, if you want the **radius** of the satellite's orbit, you must add the radius of Earth to the altitude of the satellite.



2a

(a) Centripetal force is:

- A (resultant) force that acts on / is required to keep a body moving in a circular path; [1 mark]
- That is directed towards the centre around which the body is moving / of the circle; [1 mark]

[Total: 2 marks]

2b

(b) The centripetal force when:

(i) Venus orbits the Sun is:

- Gravitational (force); [1 mark]

(ii) A car driving around a roundabout is:

- Friction / frictional (force); [1 mark]

(iii) Rotating a ball tied to the end of a piece of string:

- Tension (in the string); [1 mark]

[Total: 3 marks]

2c

(c) The path of the moon is circular because:

- The field lines show that the (gravitational) force acts towards the centre of the planet **OR** the velocity of the Moon is perpendicular to the field lines; [1 mark]
- The force is perpendicular to the velocity which causes centripetal acceleration; [1 mark]

[Total: 2 marks]

2d

(d) The distance of the Moon from the surface of the Earth is:

List the known quantities:

- Centripetal force of the Moon, $F = 2.21 \times 10^{20} \text{ N}$
- Mass of the Moon, $m = 7.35 \times 10^{22} \text{ kg}$
- Radius of Earth, $R = 6.4 \times 10^6 \text{ m}$
- Linear speed of the Moon, $v = 1023 \text{ m s}^{-1}$

Determine the equation for the radius of the Moon's orbit, r :

$$\bullet \quad F = \frac{mv^2}{r} \Rightarrow r = \frac{mv^2}{F} \quad [1 \text{ mark}]$$

Substitute in the values:

$$\bullet \quad r = \frac{(7.35 \times 10^{22}) \times (1023)^2}{2.21 \times 10^{20}} \quad [1 \text{ mark}]$$

$$\bullet \quad r = 3.4815 \times 10^8 \text{ m} \quad [1 \text{ mark}]$$

Calculate the distance from the surface of the Earth, h :

$$\bullet \quad r = R + h \Rightarrow h = r - R$$

$$\bullet \quad h = (3.4815 \times 10^8) - (6.4 \times 10^6) = 3.42 \times 10^8 \text{ m} \quad [1 \text{ mark}]$$

[Total: 4 marks]

(b) The centripetal force when:

(i) Venus orbits the Sun is:

- Gravitational (force); [1 mark]

(ii) A car driving around a roundabout is:

- Friction / frictional (force); [1 mark]

(iii) Rotating a ball tied to the end of a piece of string:

- Tension (in the string); [1 mark]

[Total: 3 marks]

2c

(c) The path of the moon is circular because:

- The field lines show that the (gravitational) force acts towards the centre of the planet **OR** the velocity of the Moon is perpendicular to the field lines; [1 mark]
- The force is perpendicular to the velocity which causes centripetal acceleration; [1 mark]

[Total: 2 marks]

2d

(d) The distance of the Moon from the surface of the Earth is:

List the known quantities:

- Centripetal force of the Moon, $F = 2.21 \times 10^{20} \text{ N}$
- Mass of the Moon, $m = 7.35 \times 10^{22} \text{ kg}$
- Radius of Earth, $R = 6.4 \times 10^6 \text{ m}$
- Linear speed of the Moon, $v = 1023 \text{ m s}^{-1}$

Determine the equation for the radius of the Moon's orbit, r .

- $F = \frac{mv^2}{r} \Rightarrow r = \frac{mv^2}{F}$ [1 mark]

Substitute in the values:

- $r = \frac{(7.35 \times 10^{22}) \times (1023)^2}{2.21 \times 10^{20}}$ [1 mark]
- $r = 3.4815 \times 10^8 \text{ m}$ [1 mark]

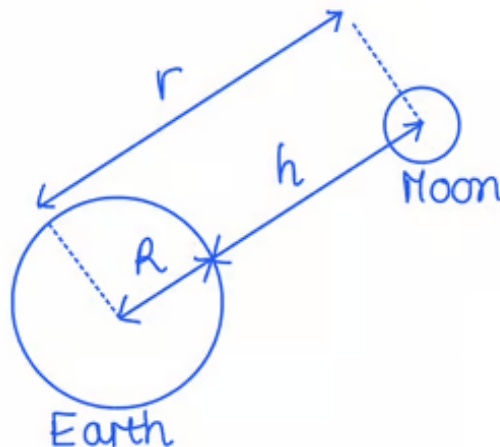
Calculate the distance from the surface of the Earth, h :

- $r = R + h \Rightarrow h = r - R$
- $h = (3.4815 \times 10^8) - (6.4 \times 10^6) = 3.42 \times 10^8 \text{ m}$ [1 mark]

[Total: 4 marks]

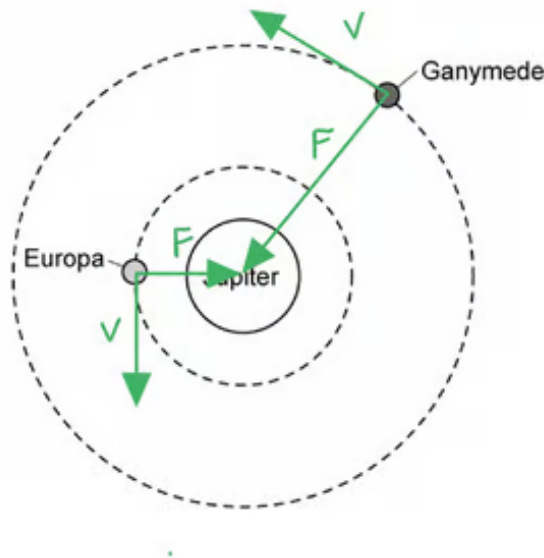
Always make sure you use the correct value for the radius of orbit, r . This is the distance from the **centre** of the two bodies.

Therefore, if you want the distance from the **surface** of the Earth to the Moon, you must subtract the radius of Earth from the value of r calculated.



3a

(a) A sketch that would score 4 marks should look like:



(i) The centripetal force on each moon is sketched as:

- Arrow drawn from the centre of each moon to the centre of Jupiter; [1 mark]
- Arrow pointed towards the centre of Jupiter **AND** labelled F ; [1 mark]

(ii) The linear velocity of each moon is sketched as:

- Arrow drawn from the centre of each moon and perpendicular to F ; [1 mark]
- Arrow pointed in the counter-clockwise direction to F **AND** labelled v ; [1 mark]

[Total: 4 marks]

3b

(b) The moon with the longer time period is:

- Ganymede; [1 mark]

Explanation:

- The radius of the orbit (r) is directly proportional to the time period (T) / $r \propto T$ **AND** as the linear velocity (v) is the same; [1 mark]
- Ganymede has a larger orbital radius (r) (so a larger time period); [1 mark]

[Total: 3 marks]

We can see Ganymede has a larger orbital radius from Fig 1.1.

We have been given their linear velocity and we want to see how the time periods

compare. The equation $v = \frac{2\pi r}{T}$ links all three of these variables.

Multiplying both sides by T gives $Tv = 2\pi r$. Since v , and 2π are constant, T and r are directly proportional.

3c

(c) The centripetal force on Ganymede is:

List the known quantities:

- Centripetal acceleration, $a = 0.11 \text{ m s}^{-2}$
- Mass, $m = 1.48 \times 10^{23} \text{ kg}$

Calculate the centripetal force, F :

- $F = ma = (1.48 \times 10^{23}) \times 0.11$ [1 mark]
- $F = 1.628 \times 10^{22} = 1.6 \times 10^{22} \text{ N}$ [1 mark]

[Total: 2 marks]

The difference between the equation for centripetal force $F = \frac{mv^2}{r}$ and centripetal acceleration $a = \frac{v^2}{r}$ is the m . They are related by Newton's second law,

$$F = ma = \frac{mv^2}{r}.$$

3d

(d) The centripetal force of an object with the same mass in the same orbit and twice the angular speed would be:

- Four times greater / quadrupled (that of Ganymede); [1 mark]

[Total: 1 mark]

The relation between the centripetal force and the angular speed is from $F = mr\omega^2$. Since the object has the same mass m and orbital radius r , we have the relation $F \propto \omega^2$. Therefore, as ω is doubled it becomes $(2\omega)^2 = 4\omega^2$. This means F becomes $4F$.